



Smartphone market decline

Gartner predicts the 'biggest ever' smartphone market decline with a drop of 3.2 percent in sales. <https://digit.in/oct19-17>



Firefox Preview 2.0 is out

The company claims that the Android browser is up to two times faster than previous versions of Firefox. <https://digit.in/oct19-18>

graphic information system) overlaid on top of these maps.

A photogrammetrist usually spends a good chunk of their work lives in their office, analysing and interpreting images to create 3D models or maps out of it, however, they can be required to extensively travel as well



LiDAR, extensively used by photogrammetrists, is based on laser beams and the measurement of the time it takes for light to return.

to the locations that are being mapped. Some photogrammetrists will have to plan aerial and satellite surveys so as to ensure complete coverage of the area they are working on.

Photogrammetrists are increasingly working on online and mobile maps, and advanced ones, such as interactive maps are becoming exceedingly popular as well. It is essential that new-age photogrammetrists learn new techniques and software to help them collect data and design maps for mobile phones and navigation systems.

QUALIFICATIONS

If you have a passion for aerial photography, geography and map-making, you may want to look into the field of photogrammetry. However, it is essential that you work to keep yourself updated with the advancements in technology. Typically, an aspiring photogrammetrist must hold a bachelor's degree in geomatics, geography, and cartography or surveying. Sometimes, individuals holding bachelor's degrees in computer science, engineering and forestry are also given the pass. It is integral that you familiarise yourself with certain tools and software utilised for image processing, light imaging detection and ranging technology (LiDAR), and remote sensing.

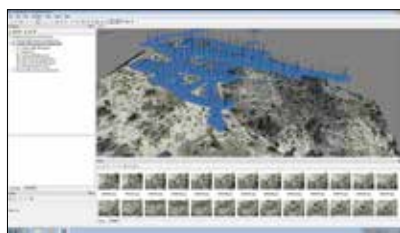
You must also have some knowledge about web-based mapping technologies, as well as new methods of compiling data that incorporate the GPS capabilities of smartphones and in-car navigation systems. The exact qualifications for becoming a photogrammetrist vary from place-to-place and country-to-country, however, experience in working with computer data, coding and critical thinking skills, as well as an affinity for creating maps, signify that you may be able to pursue a career in this field.

COMMONLY-USED SOFTWARE

Software for photogrammetry must be cautiously selected depending on your expectations. Do you need to upload multiple images? Are you focusing on aerial images? Are you a novice or experienced? Questions like these must be answered and you will be able to pick the software best-suited to your specific needs. Let's have a look at a few conventional software.

Agisoft PhotoScan

This is a stand-alone software which offers useful features such as photogrammetric triangulation, point cloud data, measurements for distances, volumes, areas, 3D model genera-



Agisoft PhotoScan

tion and textures, and more. You will have to choose between the standard and professional version. If you are a beginner or have limited needs, the standard version of Agisoft features all the basic software tools that you could possibly need which is why it is one of the most widely-used software for photogrammetry. However, if you need the ground control point option, you should upgrade to the professional version.

Reality Capture

This software is lauded for being able to process data much quicker than its competitors. It can also handle a sizeable number of images on a regular-powered desktop as long as it



Reality Capture

comes equipped with an Nvidia GPU. Its quick align feature is extremely convenient since it allows you to align images in a few seconds, and while this feature is on Agisoft as well, Reality Capture is much faster.

Pix4Dmapper

This software is primarily dedicated to drone-based mapping. It has an advanced digital reconstruction technology that can turn images into precise geo-referenced 2D maps or 3D models. You can also render your project and work on textures on this software. Pix4Dmapper is quite versatile as well since it works well on both aerial images as well as close-ranged images.

Autodesk ReCap Pro

Previously known as Autodesk Re-make, Autodesk ReCap Pro is a reality capture software that allows users to create accurate 3D models using reality captures. It lets you work with aerial photogrammetry and measure and edit point cloud data. There is an array of tools present such as clearing unwanted objects to work specifically on a precise object.

Context Capture

Context Capture assists in creating complete 3D models using photographs with no human intervention whatsoever. This software also includes the ContextCapture Editor which allows users to edit meshes, generate cross-sections and extract ground and break-lines. 